Indian Institute of Technology Hyderabad

Modern Physics End Semester Exam

Date: 27th November 2024

Full Marks: 50

Duration: 9:00 AM to 10:00 AM

Instructions:

- Each question has only one correct option.
- Circle the correct option with pen.
- No other scribbling is allowed.
- For each correct answer, 2 marks will be awarded.
- For each incorrect answer, 0.5 marks will be deducted.

Questions

1. Which of the following is a fundamental particle?
A. Electron B. Proton C. Neutron D. Atom

2. In Thomson's experiment for measuring an electron's charge-to-mass (e/m), the initial velocity of the electron is equal to:
A. Be/E B. eE C. E/B D. B/E

3. 3 MeV electrons are required to deposit a total energy of 1 J. The number required is:
A. 2×10^x B. 1.25×10^{12} C. 8×10^{13} D. 1.2×10^{19}

4. The diameter of the $^{125}_{52}$Te nuclei in fm is:
A. [illegible] B. 12 fm C. 5 fm D. 10 fm

5. The energy of the electron at which its de-Broglie wavelength is comparable to the diameter of the $^{125}_{52}$Te nuclei (Use $hc = 1240$ MeV·fm) is:
A. [illegible] B. 248 MeV C. 103.3 MeV D. 206.7 MeV

6. The correct statement about the Rutherford scattering experiment is:
A. Majority of α particles are scattered at small angles.
B. Majority of α particles are scattered at large angles (¿90°).
C. Majority of α particles are back-scattered.
D. α particles are absorbed before reaching the screen.

7. The scattering cross-section varies as $\sin^4(\theta/2)$, where θ is the scattering angle.

8. The scattering cross-section is inversely proportional to the square of the kinetic energy of the particle.

9. The amount of scattering by a given foil is approximately proportional to the (velocity)^a of the incident particles, where a is:
A. 2 B. 4 C. -2 D. -4

10. What are isotopes?
A. Nuclei with same A but different Z.
B. Nuclei with same number of neutrons.
C. Nuclei with same Z but different A.
D. Nuclei with even number of A.

11. Which particle is responsible for the missing energy and momentum in nuclear decay?
A. Neutron B. Neutrino C. α particle D. β particle

12. Choose the correct statement about quarks:
A. Do not participate in weak interactions.
B. Do not participate in strong interactions.
C. Do not participate in electromagnetic interactions.
D. Have fractional electric charges.

13. Choose the correct statement about leptons:
A. Do not participate in weak interactions.
B. Do not participate in strong interactions.
C. Have fractional electric charges.
D. Not fundamental particles.

14. Mesons are:
A. Fundamental particles
B. Made up of a quark and an anti-quark
C. Made up of two quarks
D. Made up of three quarks

15. Which of the following equations describe wave-like behavior?
A. $c = \lambda\nu$ B. $E = h\nu$ C. $\lambda = h/p$ D. $E = mc^2$

16. The de Broglie wavelength of an electron having kinetic energy of 100 eV is:
A. 1.22 nm B. 0.122 nm C. 12.20 nm D. 0.10 nm

17. If the phase velocity is the same for all frequencies and wavelengths, then which of the following is true?
A. Group velocity same as phase velocity.
B. Medium is dispersive.
C. Group velocity smaller than phase velocity.
D. Group velocity larger than phase velocity.

18. If the splitting of $2p_{3/2}$ and $2p_{1/2}$ is 45 μeV, then the magnetic field experienced

by the 2p electron is ($\mu_B = 5.79 \times 10^{-5}$ eV/T):
A. 1.17 T B. 0.78 T C. 0.39 T D. 0.19 T

19. Consider a gas in a box with thermally insulating walls (no heat exchange). The volume is suddenly changed from V_i to V_f. The change in entropy is:
A. > 0 B. < 0 C. $= 0$ D. Any of the above

20. Consider a gas in a box with thermally conducting walls (heat exchange possible). The volume is suddenly changed from V_i to V_f. The change in entropy is:
A. > 0 B. < 0 C. $= 0$ D. Any of the above

21. To describe a hydrostatic system with 1 mole of particles, I calculate entropy using $S = k_B \ln \Omega$ without using Gibbs' idea. Which of the following partial derivatives gives wrong results?
(i) $(\partial S/\partial E)_{V,N} = 1/T$
(ii) $(\partial S/\partial V)_{E,N} = P/T$
(iii) $(\partial S/\partial N)_{E,V} = -\mu/T$
A. (i) B. (i) and (ii) C. (iii) D. All are wrong

22. I have 2 distinguishable particles to put into 4 energy levels (no restriction). Number of distinct ways:
A. 10 B. 16 C. 8 D. 6

23. I have 2 indistinguishable particles to put into 4 energy levels (no restriction). Number of distinct ways:
A. 10 B. 16 C. 8 D. 6

24. I have 2 indistinguishable particles to put into 4 energy levels, not allowed to put more than one in a level. Number of distinct ways:
A. 10 B. 16 C. 8 D. 6

25. For a hydrostatic system, two entropy expressions are compared:
(i) $S = S(E, V, N)$ with $(\partial S/\partial N)_{E,V} = -\mu/T$
(ii) $S = S(\mu, T, N)$ with $(\partial S/\partial N)_{\mu,T} = -\mu/T$
Which is correct?
A. Both give $-\mu/T$ B. (i) gives, (ii) doesn't C. (ii) gives, (i) doesn't D. Neither gives